

A Baselines

The first group is graph classification models:

- GCN [8]: a GNN that uses a convolution function on a graph to propagate information within the neighborhood of nodes;
- GraphSAGE [6]: a GNN that leverages a sampling technique to aggregate features from the neighborhood.
- GAT [13]: a GNN that adopts an attention mechanism within the neighborhood of each node;
- GIN [20]: a GNN that follows graph isomorphism to capture the properties of a graph.
- LRGNN [17]: a GNN stacking multiple GNNs to extract the long-range dependencies;
- GRDL [16]: a GNN treating node embeddings as a discrete distribution, enabling direct classification without global pooling.
- BiView [3]: a GNN utilizing both homophily and heterophily for graph classification.

The second group is GAD models:

- iGAD [25]: a GNN with a substructure-aware component to capture properties of anomalous graphs.
- GmapAD [12]: a GNN mapping graphs into a latent space where anomalies can be effectively detected;
- RQGNN [5]: a GNN using Rayleigh Quotient to obtain information from both spectral and spatial spaces.
- UniGAD [10]: a GNN that unifies different levels of graph-related tasks.
- GADTrans [22]: a GNN with transformer-based backbone for graph anomaly detection.

The third group is graph-level augmentation frameworks:

- MAA [23]: a framework using node split and merge, and sub-graph mix to augment graphs heuristically;
- GLA [24]: a framework augmenting data in the representation space from the most difficult direction while keeping the label of augmented data the same as the original samples;
- GM [7]: a framework that interpolates graphons of different classes in the Euclidean space to get mixed graphons;
- FGWM [11]: a framework that seeks a midpoint of source graphs in the Fused Gromov-Wasserstein metric space to interpolate graphons of different classes.
- DP [19]: a framework that uses mask or noise to perturb the spectral spaces of graphs for augmentation.

B Algorithm

Algorithm 1: Preprocess

Input: $\mathcal{D}, k_l, k_s$

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1 for  $G$  in  $\mathcal{D}$  do
2    $G.A \leftarrow \frac{1}{2}(I + G.D^{-\frac{1}{2}} * G.A * G.D^{-\frac{1}{2}});$ 
3    $G.U_l, G.A_l \leftarrow \text{EVD}(G.A, k_l);$ 
4    $G.U_s, G.A_s \leftarrow \text{EVD}(G.A, k_s);$ 

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C Time Complexity Analysis

For the Preprocess function, we first analyze the time complexity of the matrix multiplication. Since we utilize sparse matrices to

Algorithm 2: FGG

Input: $\mathcal{D}, H_l, H_s$

Output: $\mathcal{D}'$

```

1 for  $G$  in  $\mathcal{D}$  do
2   for  $i = 0$  to  $H_l$  do
3      $G_l \leftarrow G_l + \omega_l[i] * G.U_l * G.A_l^{\alpha_l[i]} * G.U_l^T;$ 
4   for  $i = 0$  to  $H_l$  do
5      $G_s \leftarrow G_s + \omega_s[i] * G.U_s * G.A_s^{\alpha_s[i]} * G.U_s^T;$ 
6    $G' \leftarrow \omega * G_l + (1 - \omega) * G_s;$ 
7    $\mathcal{D}' \leftarrow \mathcal{D}' \cup G';$ 
8 Return  $\mathcal{D}';$ 

```

Algorithm 3: WDML

Input: $f, \mathcal{D}, \mathcal{D}'$

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1 for  $G, G'$  in  $\mathcal{D}, \mathcal{D}'$  do
2    $s, o \leftarrow f(G);$ 
3    $s', o' \leftarrow f(G');$ 
4    $m \leftarrow \frac{1 - \cos(o, o')}{2};$ 
5    $L_{\text{WDML}} \leftarrow L_{\text{WDML}} + \frac{1}{N_{G.y}} \log \frac{e^{s[G.y] - m}}{e^{s[G.y] - m} + e^{s[1 - G.y]}};$ 
6  $L_{\text{WDML}}.\text{backward}();$ 

```

Algorithm 4: MVP

Input: $f, \mathcal{D}, \mathcal{D}', \tau_n, \tau_a$

Output: $\mathcal{D}''$

```

1 for  $G, G'$  in  $\mathcal{D}, \mathcal{D}'$  do
2    $s, o \leftarrow f(G);$ 
3    $s', o' \leftarrow f(G');$ 
4   if  $s[0] < \tau_n \wedge s'[0] \leq \tau_n$  then
5      $G.y \leftarrow 0;$ 
6      $\mathcal{D}'' \leftarrow \mathcal{D}'' \cup G;$ 
7   else if  $s[1] < \tau_a \wedge s'[1] \leq \tau_a$  then
8      $G.y \leftarrow 1;$ 
9      $\mathcal{D}'' \leftarrow \mathcal{D}'' \cup G;$ 
10 Return  $\mathcal{D}'';$ 

```

Algorithm 5: FracAug

Input: $f, \mathcal{D}_{\text{train}}, \mathcal{D}_{\text{val}}, \mathcal{D}_{\text{test}}, H_l, H_s, k_l, k_s, e_{\text{warmup}}, e_{\text{aug}}, \tau_n, \tau_a$

```

1  $\text{Preprocess}(\mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{val}} \cup \mathcal{D}_{\text{test}}, k_l, k_s);$ 
2  $\mathcal{D}'_{\text{train}} \leftarrow \mathcal{D}_{\text{train}};$ 
3 for  $e = 0$  to  $e_f$  do
4   if  $e > e_{\text{warmup}} \wedge e \% e_{\text{aug}} == 0$  then
5     for  $e' = 0$  to  $e_{\text{FGG}}$  do
6        $\mathcal{D}_{\text{temp}} \leftarrow \text{FGG}(\mathcal{D}_{\text{train}}, H_l, H_s);$ 
7        $\text{WDML}(f, \mathcal{D}, \mathcal{D}_{\text{temp}});$ 
8        $\mathcal{D}_{\text{temp}} \leftarrow \text{FGG}(\mathcal{D}_{\text{val}} \cup \mathcal{D}_{\text{test}}, H_l, H_s);$ 
9        $\mathcal{D}_{\text{temp}} \leftarrow \text{MVP}(f, \mathcal{D}_{\text{val}} \cup \mathcal{D}_{\text{test}}, \mathcal{D}_{\text{temp}}, \tau_n, \tau_a);$ 
10       $\mathcal{D}'_{\text{train}} \leftarrow \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{temp}};$ 
11  $\text{train}(f, \mathcal{D}'_{\text{train}});$ 

```

Table 10: Comparison of average running time (s).

Datasets	MAAv+FA	NS	SM	GLAv+FA	GLA	GMv+FA	GM	FGWMv+FA	FGWM
MCF-7	92.18+58.56	1282.84	876.39	92.00+73.57	1493.55	90.54+107.51	51.64	92.29+106.76	553.90
MOLT-4	133.10+83.70	1352.07	881.00	128.18+113.70	1924.35	131.30+150.27	64.55	129.84+176.65	855.30
PC-3	91.26+70.05	1249.30	879.76	90.55+64.84	1484.25	90.46+110.26	51.52	90.70+98.88	613.19
SW-620	133.80+102.96	1248.39	875.52	136.75+99.17	2076.36	133.42+158.47	75.68	133.50+153.30	873.41
NCI-H23	130.20+108.73	1301.09	883.09	130.63+93.86	2091.18	131.54+134.20	63.68	131.07+143.53	896.41
OVCAR-8	131.48+98.46	1351.17	890.68	132.62+102.36	2077.41	132.07+168.57	61.41	131.23+130.47	986.01

Table 11: Comparison of average wall-clock time (s).

Datasets	MAAv+FA	NS	SM	GLAv+FA	GLA	GMv+FA	GM	FGWMv+FA	FGWM
MCF-7	135.73+69.63	1410.86	912.23	136.19+92.94	1543.64	131.46+107.54	70.60	132.19+106.70	949.81
MOLT-4	191.15+93.22	1433.10	936.19	194.52+127.01	1993.29	189.00+155.83	98.79	188.37+164.47	1487.41
PC-3	133.72+74.67	1423.05	973.39	132.97+86.91	1524.84	131.72+116.66	68.41	131.70+103.64	968.09
SW-620	193.47+104.12	1411.45	945.87	191.01+130.31	2138.65	194.62+162.24	102.90	191.97+162.04	1321.40
NCI-H23	191.622+113.72	1417.68	896.13	191.50+118.83	2220.64	192.70+139.89	102.48	194.30+150.65	1438.07
OVCAR-8	194.69+102.93	1459.6	891.68	192.11+127.31	2521.36	190.40+160.44	102.89	193.93+171.23	1430.77
P388	171.61+103.73	1573.25	933.75	172.97+130.55	2146.63	171.64+146.75	104.16	175.47+195.82	1754.24
SF-295	193.68+93.42	1640.00	889.18	195.20+129.01	2065.14	190.63+165.59	102.69	194.14+185.80	1469.35
SN12C	191.82+102.57	1429.26	972.91	192.14+125.52	2116.96	192.33+145.00	101.46	192.01+185.39	1378.80
UACC257	191.56+100.38	1429.63	977.20	193.70+122.99	2110.75	192.43+159.39	102.32	189.07+164.62	1320.10
PROTEINS_full	12.90+7.600	213.95	216.55	12.28+8.87	80.33	12.91+9.25	6.65	12.04+8.55	42.86
DBLP_v1	69.62+67.53	2238.69	1866.97	66.97+74.43	3155.02	69.79+99.96	61.35	66.85+104.41	227.15

conduct the experiment, the time complexity of the multiplication is $O(\text{nnz}(G.D^{-\frac{1}{2}}) * \text{nnz}(G.A) + \text{nnz}(G.D^{-\frac{1}{2}} * G.A) * \text{nnz}(G.D^{-\frac{1}{2}}))$, where nnz means non-zero entries of the matrix. Then, by adopting EVD to only keep the top- k_l largest and top- k_s smallest eigenvalues, the time complexity can be $O(n * (k_l^2 + k_s^2) + m * (k_l + k_s))$, where n, m is the number of nodes/edges. As shown in Algorithm 5, Preprocess can be called before the training process, and thus it won't burden the training or inference of our FracAug.

Then, we analyze the time complexity of FGG for each graph. As presented in Algorithm 2, we perform sparse matrix multiplication for every sample H_l and H_s times. Besides, since $G.A$ is a diagonal matrix, the time complexity of multiplying $G.A$ is the same as that of multiplying $G.A^a$. Therefore, the total time complexity of FGG is $O(\text{FGG}) = O(\text{nnz}(G.U_l) * k_l + \text{nnz}(G.U_l * G.A_l) * \text{nnz}(G.U_l^T) + \text{nnz}(G.U_s) * k_s + \text{nnz}(G.U_s * G.A_s) * \text{nnz}(G.U_s^T))$.

Next, we analyze the time complexity of WDML in Algorithm 3. Assuming that we only have one sample in $\mathcal{D}_{train}$, then the time complexity of WDML is $O(\text{WDML}) = O(d)$, where d is the dimension of the generated graph embedding $\mathbf{o}$.

Moreover, as shown in Algorithm 4, in MVP, we only need to see if the probability predicted by the given GNN satisfies the criterion, so the time complexity for MVP is $O(\text{MVP}) = O(1)$.

Finally, in Algorithm 5, we combine all the time complexities together within one training epoch of the given GNN f , assuming the time complexity of f for each sample is $O(f)$, then we have the total complexity as $O(e_{\text{FGG}} * (O(\text{FGG}) + O(\text{WDML}) + O(f)) * N_{train} + (O(\text{FGG}) + O(\text{MVP}) + O(f)) * (N_{val} + N_{test}))$, where N_{val}, N_{test} represent the number of samples in $\mathcal{D}_{val}$ and $\mathcal{D}_{test}$, respectively.

In practice, we set e_{FGG} to 10 and e_{aug} to 25, which can reduce the computational cost, and FGG can still converge. According to the final complexity, we can see the dominant factor within each epoch is $O(e_{\text{FGG}} * O(f) * N_{train} + O(f) * (N_{val} + N_{test}))$. For such

a factor, we need to calculate it in total $\frac{ef - e_{\text{warmup}}}{e_{\text{aug}}} * e_{\text{FGG}}$ times, which is much less than the original training epoch of f . Hence, the increase in time complexity will not be the limitation of our FracAug in real applications.

In Tables 10 and 11, we present a detailed runtime comparison and wall-clock time comparison between FracAug and several leading graph augmentation methods, respectively. For each technique, we decompose the total computational cost into a one-time preprocessing phase, performed once per dataset, and the subsequent training time measured over multiple epochs. While some baselines require repeated feature perturbations or costly online sampling at every iteration, FracAug's eigenvalue decomposition is only performed during preprocessing. As a result, the per-epoch training overhead of FracAug remains on par with, or even below, that of competing approaches, despite leveraging additional spectral information to boost anomaly detection performance.

Crucially, these efficiency gains do not come at the expense of detection performance. Across all datasets and baseline comparisons, FracAug consistently delivers state-of-the-art AUROC, AUPRC, and F1-score results while maintaining competitive total runtimes. By amortizing the heavier spectral computations over the entire training cycle and by implementing optimized matrix operations, FracAug strikes an effective balance between computational tractability and augmentation quality. This combination of speed and performance underscores the practical value of our method: practitioners can readily adopt FracAug for real graph applications without incurring prohibitive time costs.

D Memory Complexity Analysis

In Tables 12 and 13, we report detailed comparisons of total and peak memory consumption for FracAug against several leading graph augmentation methods. For each method in Table 7, we further

Table 12: Comparison of average consumed memory (MB).

Datasets	MAAv+FA	NS	SM	GLAv+FA	GLA	GMv+FA	GM	FGWMv+FA	FGWM
MCF-7	171.99+530.47	655.12	689.79	171.88+687.56	1125.73	172.20+538.38	675.25	172.32+552.62	598.73
MOLT-4	240.71+534.24	763.71	784.25	239.91+672.44	1262.11	239.92+538.74	770.23	239.85+573.34	817.38
PC-3	170.76+530.65	652.03	687.16	170.68+661.78	1118.31	170.50+533.72	671.04	170.77+505.36	602.24
SW-620	243.48+530.75	764.33	801.05	241.11+666.00	1276.12	243.46+531.57	769.41	249.02+546.39	838.00
NCI-H23	248.02+531.50	761.09	798.24	247.96+678.73	1300.57	250.67+534.30	772.79	247.66+549.04	836.02
OVCAR-8	248.82+526.89	764.66	800.23	248.93+669.54	1247.54	243.27+536.46	771.64	243.43+550.80	805.35

Table 13: Comparison of average peak memory (MB).

Datasets	MAAv+FA	NS	SM	GLAv+FA	GLA	GMv+FA	GM	FGWMv+FA	FGWM
MCF-7	62.53	55.84	52.07	73.49	143.10	60.55	51.83	65.46	137.32
MOLT-4	89.45	78.66	73.17	95.55	192.97	86.34	79.51	90.78	194.61
PC-3	61.93	54.51	51.37	71.91	142.55	60.67	51.40	72.49	137.59
SW-620	91.19	80.08	74.80	85.14	196.15	93.25	81.22	93.69	197.01
NCI-H23	90.79	79.15	74.23	99.11	195.42	87.77	81.13	96.70	197.96
OVCAR-8	91.16	80.31	74.50	97.86	196.09	92.59	81.50	94.43	195.91
P388	93.28	82.13	76.83	103.88	200.72	95.81	90.00	96.29	208.33
SF-295	90.61	78.46	74.34	94.73	195.08	86.58	81.05	95.56	196.82
SN12C	90.02	78.43	73.81	96.41	193.95	90.44	80.36	94.84	199.04
UACC257	89.98	79.26	74.08	96.52	193.88	88.65	80.81	90.62	194.10
PROTEINS_full	4.89	5.83	5.63	4.97	6.51	5.01	7.02	4.95	7.35
DBLP_v1	45.35	38.40	37.67	48.54	95.30	43.37	29.05	46.59	93.41

Table 14: Hyperparameters of 12 datasets based on GIN.

Datasets	MCF-7	MOLT-4	PC-3	SW-620	NCI-H23	OVCAR-8	P388	SF-295	SN12C	UACC257	PROTEINS_full	DBLP_v1
k_l	4	4	4	4	3	3	4	3	4	4	3	4
H_l	4	3	3	3	3	3	4	4	3	4	4	4
k_s	3	4	3	3	4	3	4	3	3	4	3	3
H_s	4	4	3	3	3	3	4	3	3	4	4	3
e_{warmup}	50	25	50	50	25	25	50	50	25	50	50	25

break down the total memory usage into a one-time preprocessing stage, executed once per dataset, and the recurring memory cost measured across training epochs. For each method in Table 13, we present the peak memory during the training procedure. Unlike baselines that repeatedly perturb features or perform expensive online sampling at every iteration, FracAug requires eigenvalue decomposition only during preprocessing. Consequently, its per-epoch training overhead remains comparable to, or even lower than, that of competing methods, despite incorporating additional spectral information to enhance anomaly detection.

Importantly, these memory efficiency benefits do not compromise detection quality. Across all datasets and baselines, FracAug consistently achieves state-of-the-art AUROC, AUPRC, and F1-scores while maintaining competitive end-to-end memory cost. By amortizing the heavier spectral computations across the full training process and employing optimized matrix operations, FracAug achieves a strong balance between memory efficiency and augmentation quality. This blend of memory usage and performance highlights the method’s practical value: practitioners can deploy FracAug in real-world graph applications without incurring prohibitive memory costs.

E Experimental Settings

Table 14 provides a comprehensive summary of the hyperparameter configurations used in our experiments based on the GIN backbone. To obtain the final settings for FracAug, we perform a grid search and select the configuration that achieves the highest combined AUROC, AUPRC, and F1-score on the validation set, after which we report the corresponding test performance. Concretely, the parameters k_l, H_l, k_s, H_s are each chosen from the set 3, 4, and the number of GNN warmup epochs is selected from 25, 50, which helps keep the overall search cost manageable while still ensuring sufficient coverage of plausible configurations. For fairness and reproducibility, all experiments are conducted on an NVIDIA Quadro RTX 8000, ensuring a consistent computational environment across all evaluations.

F Margin Loss Comparison

Next, we examine the performance of FracAug under different margin losses, as discussed in Section 4.3, by conducting comparisons across 12 datasets using the GIN backbone. As reported in Table 15, FracAug consistently surpasses its variants that employ Softmax, LMCL, or LDAM as alternative margin formulations. This consistent advantage highlights the effectiveness of our design and

Table 15: Comparison of different margin loss.

Datasets	Metrics	GIN	+FA	Softmax	LMCL	LDAM
MCF-7	AUC	0.5867	0.5976	0.5844	0.5880	0.5844
	AUPRC	0.2830	0.2971	0.2781	0.2847	0.2796
	MF1	0.5366	0.5421	0.5378	0.5372	0.5360
MOLT-4	AUC	0.5733	0.5854	0.5760	0.5797	0.5751
	AUPRC	0.2830	0.3001	0.2969	0.2952	0.2857
	MF1	0.5072	0.5103	0.4991	0.5059	0.5076
PC-3	AUC	0.5969	0.6119	0.6021	0.6037	0.6001
	AUPRC	0.2797	0.2893	0.2809	0.2778	0.2769
	MF1	0.5063	0.5205	0.5134	0.5195	0.5144
SW-620	AUC	0.5938	0.6004	0.5947	0.5936	0.5931
	AUPRC	0.2776	0.2813	0.2779	0.2768	0.2720
	MF1	0.5090	0.5155	0.5100	0.5092	0.5133
NCI-H23	AUC	0.5897	0.5968	0.5913	0.5896	0.5887
	AUPRC	0.2566	0.2659	0.2650	0.2612	0.2622
	MF1	0.5059	0.5073	0.5001	0.5012	0.4990
OVCAR-8	AUC	0.5935	0.5963	0.5905	0.5890	0.5932
	AUPRC	0.2573	0.2612	0.2557	0.2570	0.2579
	MF1	0.5118	0.5123	0.5087	0.5051	0.5107

reinforces the idea that sample-specific decision boundaries provide a more suitable and flexible mechanism than fixed margins for graph-level anomaly detection. The results align closely with our theoretical analysis in Section 4.3, further validating the motivation and significance of adopting adaptive, distance-aware margins within the FracAug framework.

G Performance with More Training Data

To further demonstrate the superior capability of our proposed FracAug framework, we conduct additional experiments on PROTEINS_full and DBLP_v1 under varying training sizes (%), as illustrated in Figures 5 and 6. Across all training-size configurations, the red curves, representing the performance of GIN equipped with FracAug, consistently appear at the top of the plots. This clear and repeated trend shows that as more training data becomes available, FracAug continues to deliver steady performance gains over the baseline in terms of AUROC, AUPRC, and F1-score. Overall, these results provide additional evidence that FracAug offers a robust, broadly applicable, and practical enhancement to existing models, maintaining its advantages even as the amount of supervised data varies.

In addition, we examine the inductive setting, where only the validation set is used for pseudo-labeling, to evaluate the generalization capabilities of our proposed framework, FracAug. This analysis is represented by the yellow curves in Figures 5 and 6. Across all training-size configurations, the yellow curves, which depict the performance of the GIN model enhanced with FracAug under this setting, consistently outperform the vanilla model. While the yellow curve falls slightly below the red curve due to the reduced amount of data available, the performance gap is minimal. This further underscores the viability of our proposed framework in strictly inductive scenarios. Overall, these findings highlight that FracAug provides a robust, versatile, and practical improvement to existing models, maintaining its effectiveness even when limited to the validation set for pseudo-labeling.

H Learned Parameters

Here, we present the learned parameters based on GIN, α_l , ω_l , α_s , ω_s , and ω , as summarized in Table 16. Specifically, α_l and α_s correspond to the learned fractional powers associated with the largest and smallest eigenvalue groups, respectively, while ω_l and ω_s denote the learned coefficients applied to the fractional graphs generated from these eigenvalues. The vector ω represents the balancing coefficient between the two groups of fractional graphs derived from the largest and smallest eigenvalues.

As described in Section 4.2, the hyperparameters H_l and H_s determine the number of fractional graphs combined within each eigenvalue group. Consequently, the dimensionality of α_l and ω_l equals H_l , and that of α_s and ω_s equals H_s . Since we only consider two eigenvalue groups, the largest and the smallest, the vector ω always contains two entries. Moreover, as indicated by Table 14, the optimal values of H_l and H_s vary across datasets, leading to corresponding variations in the number of entries reported in Table 16. In cases where a particular entry does not exist due to the selection of H_l or H_s , we mark it with “–” for clarity.

From Table 16, we observe that the learned fractional powers fall within the range (0, 3), which is consistent with common practices in GNN design: excessively large effective depths can cause over-smoothing, whereas overly small depths may lead to under-fitting. This observation further validates the rationality of our proposed FracAug, demonstrating that the learned parameters remain well-aligned with established principles while effectively capturing dataset-specific structural information.

I Additional Experimental Results

We present the additional experimental results for MCF-7, MOLT-4, PC-3, SW-620, NCI-H23, and OVCAR-8 to further demonstrate our superiority over other baselines. On one hand, as shown in Tables 17 and 18, FracAug consistently boosts the performance of a variety of backbone models in both the graph classification and graph-level anomaly detection methods. This pattern holds across all the newly included datasets, reinforcing the generalization ability of our proposed framework and showing that its benefits are not restricted to specific architectures or data distributions. On the other hand, as shown in Table 19, FracAug also outperforms other existing graph augmentation techniques, even when those techniques are evaluated alongside their own native backbones. This result highlights the effectiveness and robustness of our framework, indicating that FracAug can provide complementary advantages regardless of the underlying model design or augmentation strategy used.

J Standard Deviation for Experimental Results

Due to the limited space, we present the standard deviation for our experimental results here in Tables 20, 21, and 22. Across the three tables reporting standard deviations, for graph classification, graph-level anomaly detection, and data augmentation, we observe a consistent stabilizing effect introduced by FracAug. Although FracAug serves as a plug-in framework that can be seamlessly integrated with diverse baselines, its impact is both clear and substantial. In most cases, the incorporation of FracAug not only improves AUROC, AUPRC, and F1-score, as shown in Tables 17, 18, and 19, but also reduces the variance of these metrics, as shown in Tables

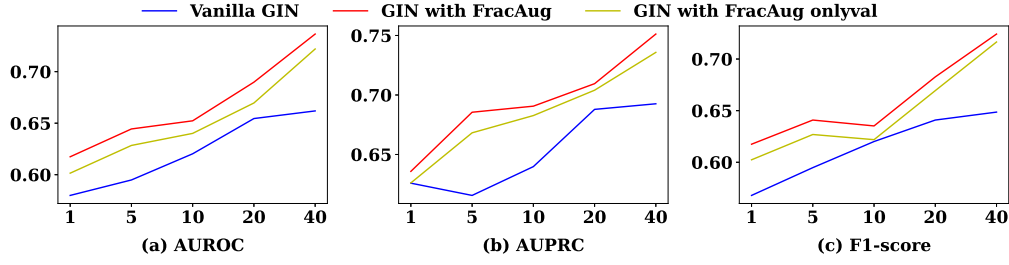


Figure 5: Varying training size (%) for PROTEINS_full.

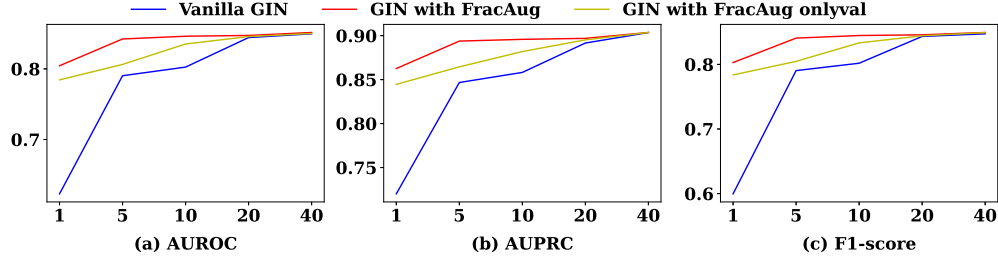


Figure 6: Varying training size (%) for DBLP_v1.

Table 16: Learned parameters

Datasets	MCF-7	MOLT-4	PC-3	SW-620	NCI-H23	OVCA8-8	P388	SF-295	SN12C	UACC257	PROTEINS_full	DBLP_v1
α_l	0.9062	1.4187	1.0862	2.186	0.4991	0.8986	2.8806	0.8845	2.7404	1.6178	1.1407	1.8663
	2.0063	2.2813	1.9282	1.5108	0.3535	0.1262	2.9620	2.9038	0.5877	1.9339	0.6885	1.1374
	1.4770	1.7911	1.9928	1.8073	2.3672	1.1480	2.3129	2.3939	1.0219	2.4874	2.5432	1.1843
	1.4269	-	-	-	-	-	1.2674	2.4786	-	2.2634	1.4364	2.8254
ω_l	0.2921	0.2655	0.3450	0.2667	0.2793	0.3476	0.1639	0.3051	0.2385	0.1165	0.3842	0.1685
	0.3108	0.3101	0.3349	0.5068	0.4612	0.3615	0.3654	0.1747	0.3833	0.2806	0.1535	0.1714
	0.1549	0.4244	0.3202	0.2266	0.2595	0.2909	0.2728	0.2145	0.3782	0.3028	0.2927	0.4004
	0.2422	-	-	-	-	-	0.1979	0.3057	-	0.3000	0.1696	0.2597
α_s	2.3128	1.9621	2.8048	1.5579	2.4828	0.5030	1.5344	1.6912	2.9331	0.8959	1.1290	1.7201
	2.1539	0.4087	1.1722	1.8088	2.1366	1.4235	1.552	1.9896	1.5150	1.9589	0.1467	2.8101
	0.7485	2.9168	1.6098	0.4474	2.2668	2.6830	2.0318	1.8741	2.9352	2.8844	0.5492	2.0935
	2.5362	0.2449	-	-	-	-	2.7135	-	-	0.1148	0.0124	-
ω_s	0.3042	0.2821	0.3910	0.3756	0.3292	0.3384	0.1696	0.4052	0.4554	0.2916	0.2407	0.2517
	0.2569	0.2112	0.2750	0.4245	0.3630	0.3436	0.4001	0.2386	0.2343	0.2539	0.3285	0.4758
	0.2764	0.1519	0.3340	0.1999	0.3078	0.3180	0.2027	0.3562	0.3103	0.3122	0.2524	0.2725
	0.1625	0.3548	-	-	-	-	0.2277	-	-	0.1423	0.1785	-
ω	0.4634	0.6795	0.6481	0.5579	0.5263	0.4541	0.4408	0.5589	0.5724	0.5309	0.7134	0.5174
	0.5366	0.3205	0.3519	0.4421	0.4737	0.5459	0.5592	0.4411	0.4276	0.4691	0.2866	0.4826

20, 21, and 22. This dual benefit indicates that FracAug enhances both the effectiveness and the reliability of downstream models. By pseudo-labeling the semantic-preserving graphs, FracAug helps models converge to more stable decision boundaries. Consequently, the tables collectively demonstrate that FracAug is not just a performance booster but also a variance reducer, offering practitioners a method that delivers stronger and more consistent results across a wide range of graph-level anomaly detection tasks.

K Additional Ablation Study

Beyond the 6 datasets examined in Section 5.3, we further conduct an ablation study on the remaining 6 datasets, as reported in Table 23. The notations used follow the same definitions as those in Table 4. In this extended analysis, we again observe that FracAug

consistently outperforms all four of its variants across the newly included datasets. This repeated pattern reinforces the importance of the individual components that constitute our framework and provides additional evidence that each design choice contributes meaningfully to the overall performance improvements achieved by FracAug.

L MVP for Validation Set Only

Although we follow the standard data augmentation setting—using both the validation and test sets for pseudo-labeling—as adopted in related works such as ConsisGAD [2], we additionally evaluate a more restrictive setting in which only the validation set is used for pseudo-labeling. This complementary experiment allows us

Table 17: Average AUROC, AUPRC, and F1-score on 6 datasets with multiple runs, using graph classification models as baselines, where the white columns represent vanilla models and the "+FA" represent models augmented by FracAug.

Datasets	Metrics	GCN +FA	SAGE +FA	GAT +FA	GIN +FA	LRGNN +FA	GRDL +FA	BiView +FA
MCF-7	AUROC	0.5753 0.5840	0.5801 0.5930	0.5885 0.6058	0.5867 0.5976	0.5467 0.6117	0.5867 0.6197	0.5602 0.6011
	AUPRC	0.3035 0.3073	0.3316 0.3340	0.3678 0.3814	0.2830 0.2971	0.2951 0.3309	0.3038 0.3469	0.2537 0.2864
	F1-score	0.4982 0.5074	0.4798 0.4963	0.4573 0.4687	0.5366 0.5421	0.4660 0.5290	0.5147 0.5254	0.5731 0.5785
MOLT-4	AUROC	0.5531 0.5650	0.5326 0.5727	0.5403 0.5721	0.5733 0.5854	0.5495 0.5851	0.5858 0.5924	0.5625 0.5791
	AUPRC	0.3183 0.3217	0.1907 0.2602	0.3456 0.3667	0.2830 0.3001	0.3047 0.3175	0.3018 0.3101	0.2378 0.2504
	F1-score	0.4501 0.4626	0.5166 0.5277	0.4096 0.4313	0.5072 0.5103	0.4570 0.4939	0.5091 0.5117	0.5707 0.5720
PC-3	AUROC	0.5697 0.5863	0.5986 0.6119	0.5707 0.5865	0.5969 0.6119	0.5690 0.6102	0.6044 0.6202	0.5664 0.6157
	AUPRC	0.2740 0.2837	0.3154 0.3248	0.3562 0.3626	0.2797 0.2893	0.2320 0.3038	0.3381 0.3459	0.2386 0.2685
	F1-score	0.4745 0.4876	0.4751 0.4841	0.4036 0.4166	0.5063 0.5205	0.5103 0.5024	0.4615 0.4750	0.5811 0.5791
SW-620	AUROC	0.5662 0.5839	0.5800 0.5968	0.5633 0.5870	0.5938 0.6004	0.5758 0.5946	0.6005 0.6046	0.5377 0.6037
	AUPRC	0.3134 0.3229	0.3401 0.3260	0.2481 0.2587	0.2776 0.2813	0.3506 0.3386	0.2908 0.2901	0.2012 0.2564
	F1-score	0.4406 0.4541	0.4339 0.4678	0.4923 0.5187	0.5090 0.5155	0.4190 0.4536	0.5059 0.5135	0.5496 0.5643
NCI-H23	AUROC	0.5811 0.5864	0.5765 0.6105	0.5777 0.6084	0.5897 0.5968	0.6002 0.6315	0.6161 0.6271	0.5600 0.5655
	AUPRC	0.2777 0.2777	0.3197 0.3207	0.2966 0.2945	0.2566 0.2659	0.2830 0.3205	0.3034 0.3069	0.2125 0.2187
	F1-score	0.4751 0.4819	0.4333 0.4740	0.4548 0.4961	0.5059 0.5073	0.4987 0.5055	0.4983 0.5112	0.5723 0.5768
OVCAR-8	AUROC	0.5692 0.5809	0.5763 0.5836	0.5396 0.5784	0.5935 0.5963	0.5628 0.5984	0.6230 0.6311	0.5417 0.5420
	AUPRC	0.3240 0.3263	0.3391 0.3423	0.2010 0.2401	0.2573 0.2612	0.3106 0.3290	0.3042 0.3129	0.2075 0.2174
	F1-score	0.4216 0.4334	0.4163 0.4221	0.4855 0.5060	0.5118 0.5123	0.4260 0.4518	0.5087 0.5119	0.5563 0.5575

Table 18: Average AUROC, AUPRC, and F1-score on 6 datasets with multiple runs, using GAD models as baselines, where the white columns represent vanilla models and the "+FA" represent models augmented by FracAug.

Datasets	Metrics	iGAD +FA	GmapAD +FA	RQGNN +FA	UniGAD +FA	GADTrans +FA
MCF-7	AUROC	0.5670 0.5756	0.5159 0.5342	0.5522 0.5709	0.5380 0.5480	0.5712 0.5856
	AUPRC	0.3402 0.3525	0.3521 0.3577	0.2379 0.2648	0.2405 0.2527	0.3029 0.3288
	F1-score	0.4546 0.4549	0.3776 0.3961	0.5609 0.5804	0.4959 0.5008	0.4926 0.4901
MOLT-4	AUROC	0.5562 0.5573	0.5100 0.5374	0.5576 0.5696	0.5353 0.5445	0.5605 0.5635
	AUPRC	0.3088 0.3092	0.2704 0.2827	0.2302 0.2468	0.2111 0.2241	0.3194 0.3221
	F1-score	0.4621 0.4636	0.4354 0.4592	0.5644 0.5718	0.5079 0.5125	0.4586 0.4602
PC-3	AUROC	0.5526 0.5674	0.5112 0.5266	0.5618 0.6043	0.5496 0.5559	0.5743 0.6007
	AUPRC	0.1887 0.2159	0.2999 0.3094	0.2370 0.2663	0.3875 0.3889	0.3999 0.3968
	F1-score	0.5217 0.5229	0.3845 0.3938	0.5721 0.5972	0.3461 0.3542	0.3646 0.4012
SW-620	AUROC	0.5641 0.5776	0.5279 0.5362	0.5428 0.5692	0.5427 0.5688	0.5658 0.5806
	AUPRC	0.3332 0.3701	0.3506 0.3479	0.1936 0.2219	0.2527 0.2585	0.3503 0.3545
	F1-score	0.4207 0.4029	0.3605 0.3734	0.5530 0.5654	0.4600 0.4903	0.4072 0.4214
NCI-H23	AUROC	0.5689 0.5721	0.5289 0.5489	0.5704 0.6061	0.5694 0.5860	0.5821 0.5888
	AUPRC	0.2267 0.2288	0.3274 0.3401	0.2166 0.2500	0.2733 0.2756	0.3368 0.3375
	F1-score	0.5039 0.5066	0.3710 0.3827	0.5770 0.5820	0.4645 0.4831	0.4247 0.4321
OVCAR-8	AUROC	0.5609 0.5685	0.5209 0.5243	0.5549 0.5773	0.5360 0.5445	0.5813 0.5824
	AUPRC	0.2319 0.2325	0.2863 0.2836	0.1933 0.2216	0.3196 0.3112	0.3231 0.3242
	F1-score	0.4880 0.4991	0.3992 0.4054	0.5618 0.5794	0.3873 0.4043	0.4365 0.4369

Table 19: Average AUROC, AUPRC, and F1-score on 6 datasets with multiple runs, using graph-level augmentation models as baselines, where the white columns represent vanilla models and their own augmentation method, while the "+FA" represent vanilla models augmented by FracAug.

Datasets	Metrics	MAAv	NS	SM	+FA	GLAv	GLA	+FA	GMv	GM	+FA	FGWMv	FGWM	+FA	DPv	DP	+FA
MCF-7	AUROC	0.5496	0.5727	0.5428	0.5695	0.5797	0.5735	0.6068	0.5730	0.5581	0.5935	0.5731	0.5502	0.5828	0.5360	0.5045	0.5897
	AUPRC	0.2346	0.2445	0.2224	0.2455	0.2558	0.2456	0.2944	0.2767	0.2864	0.3081	0.2720	0.2130	0.2945	0.1837	0.1279	0.2861
	F1-score	0.5179	0.5635	0.5512	0.5721	0.5607	0.5609	0.5793	0.5186	0.4879	0.5220	0.5585	0.5283	0.5971	0.5369	0.4923	0.5391
MOLT-4	AUROC	0.5506	0.5391	0.5113	0.5663	0.5585	0.5578	0.5792	0.5637	0.5547	0.5771	0.5477	0.5308	0.6067	0.5640	0.5755	0.5940
	AUPRC	0.2203	0.1914	0.1796	0.2356	0.2177	0.2159	0.2511	0.2411	0.2176	0.2559	0.1994	0.1939	0.3104	0.2446	0.2557	0.2896
	F1-score	0.5595	0.5392	0.5048	0.5658	0.5540	0.5519	0.5733	0.5306	0.5368	0.5416	0.5446	0.5115	0.5366	0.5275	0.5281	0.5361
PC-3	AUROC	0.5688	0.5805	0.5284	0.5821	0.6207	0.5938	0.6223	0.5705	0.5646	0.5782	0.5490	0.5442	0.6107	0.5592	0.5710	0.6011
	AUPRC	0.2112	0.2233	0.1854	0.2385	0.2770	0.2386	0.2790	0.3506	0.3456	0.3587	0.2028	0.2108	0.2623	0.2542	0.2531	0.2765
	F1-score	0.5698	0.5685	0.5321	0.5848	0.5650	0.5569	0.5686	0.4085	0.4058	0.4101	0.5027	0.4914	0.5633	0.4775	0.4852	0.5165
SW-620	AUROC	0.5577	0.5776	0.5232	0.5834	0.5822	0.5799	0.5936	0.5839	0.5722	0.5987	0.5265	0.5395	0.6183	0.5802	0.5903	0.6127
	AUPRC	0.2026	0.2205	0.1958	0.2279	0.2315	0.2258	0.2455	0.2599	0.2511	0.2728	0.1340	0.1777	0.2759	0.2578	0.2776	0.3035
	F1-score	0.5641	0.5477	0.5273	0.5676	0.5422	0.5473	0.5786	0.5111	0.5015	0.5220	0.5283	0.5113	0.5729	0.5072	0.5002	0.5119
NCI-H23	AUROC	0.5792	0.5634	0.5323	0.5979	0.5773	0.5762	0.6194	0.5912	0.5647	0.6014	0.5658	0.5760	0.6422	0.5851	0.5804	0.5929
	AUPRC	0.2273	0.1983	0.2017	0.2407	0.2082	0.2188	0.2680	0.2587	0.2139	0.2672	0.1986	0.2345	0.3105	0.2340	0.2333	0.2355
	F1-score	0.5821	0.5625	0.5445	0.5835	0.5659	0.5779	0.5914	0.5058	0.5087	0.5134	0.5561	0.5066	0.5358	0.5222	0.5225	0.5366
OVCAR-8	AUROC	0.5507	0.5494	0.5278	0.5726	0.5911	0.5859	0.6049	0.5786	0.5725	0.6024	0.5696	0.5713	0.6317	0.5808	0.5851	0.5943
	AUPRC	0.1775	0.1633	0.1665	0.2132	0.2346	0.2196	0.2447	0.2764	0.2994	0.3072	0.2696	0.2401	0.3060	0.2393	0.2405	0.2449
	F1-score	0.5461	0.5408	0.5337	0.5749	0.5350	0.5548	0.5728	0.4733	0.4469	0.4766	0.4831	0.4950	0.5211	0.5099	0.5108	0.5277

Table 20: Standard Deviation of AUROC, AUPRC, and F1-score for Table 17.

Datasets	Metrics	GCN +FA	SAGE +FA	GAT +FA	GIN +FA	LRGNN +FA	GRDL +FA
MCF-7	AUROC	0.0027 0.0016	0.0011 0.0123	0.0144 0.0011	0.0015 0.0010	0.0011 0.0069	0.0128 0.0050
	AUPRC	0.0110 0.0016	0.0024 0.0050	0.0436 0.0465	0.0036 0.0030	0.0245 0.0135	0.0256 0.0007
	F1-score	0.0136 0.0038	0.0006 0.0134	0.0215 0.0442	0.0012 0.0017	0.0182 0.0030	0.0041 0.0081
MOLT-4	AUROC	0.0080 0.0015	0.0233 0.0035	0.0028 0.0101	0.0043 0.0006	0.0122 0.0055	0.0076 0.0035
	AUPRC	0.0093 0.0032	0.0391 0.0059	0.0217 0.0024	0.0126 0.0052	0.0018 0.0107	0.0076 0.0037
	F1-score	0.0023 0.0047	0.0116 0.0003	0.0233 0.0155	0.0047 0.0058	0.0171 0.0177	0.0045 0.0092
PC-3	AUROC	0.0132 0.0047	0.0093 0.0018	0.0284 0.0150	0.0015 0.0042	0.0117 0.0015	0.0163 0.0045
	AUPRC	0.0067 0.0027	0.0092 0.0043	0.0235 0.0269	0.0017 0.0049	0.0465 0.0093	0.0332 0.0004
	F1-score	0.0109 0.0038	0.0037 0.0016	0.0124 0.0071	0.0040 0.0017	0.0258 0.0071	0.0098 0.0064
SW-620	AUROC	0.0063 0.0010	0.0008 0.0014	0.0216 0.0112	0.0038 0.0026	0.0054 0.0153	0.0087 0.0077
	AUPRC	0.0190 0.0035	0.0078 0.0045	0.0115 0.0040	0.0027 0.0001	0.0068 0.0106	0.0064 0.0004
	F1-score	0.0088 0.0045	0.0082 0.0022	0.0196 0.0237	0.0086 0.0045	0.0130 0.0292	0.0068 0.0129
NCI-H23	AUROC	0.0030 0.0033	0.0013 0.0016	0.0062 0.0020	0.0016 0.0045	0.0306 0.0085	0.0075 0.0087
	AUPRC	0.0074 0.0015	0.0031 0.0006	0.0098 0.0074	0.0014 0.0071	0.0038 0.0197	0.0061 0.0136
	F1-score	0.0026 0.0029	0.0043 0.0028	0.0009 0.0100	0.0037 0.0004	0.0411 0.0349	0.0048 0.0011
OVCAR-8	AUROC	0.0047 0.0008	0.0026 0.0010	0.0201 0.0061	0.0018 0.0014	0.0130 0.0008	0.0133 0.0009
	AUPRC	0.0017 0.0009	0.0037 0.0045	0.0220 0.0062	0.0033 0.0016	0.0061 0.0122	0.0052 0.0018
	F1-score	0.0070 0.0002	0.0003 0.0030	0.0086 0.0153	0.0005 0.0006	0.0203 0.0101	0.0037 0.0036

Table 21: Standard Deviation of AUROC, AUPRC, and F1-score for Table 18.

Datasets	Metrics	iGAD +FA	GmapAD +FA	RQGNN +FA	UniGAD +FA
MCF-7	AUROC	0.0020 0.0022	0.0006 0.0035	0.0167 0.0136	0.0069 0.0055
	AUPRC	0.0118 0.0016	0.0135 0.0086	0.0047 0.0042	0.0100 0.0025
	F1-score	0.0078 0.0015	0.0138 0.0037	0.0136 0.0067	0.0165 0.0097
MOLT-4	AUROC	0.0033 0.0020	0.0021 0.0038	0.0041 0.0132	0.0011 0.0001
	AUPRC	0.0075 0.0009	0.0168 0.0082	0.0247 0.0047	0.0226 0.0009
	F1-score	0.0105 0.0018	0.0100 0.0016	0.0122 0.0001	0.0139 0.0009
PC-3	AUROC	0.0246 0.0037	0.0096 0.0021	0.0264 0.0017	0.0040 0.0009
	AUPRC	0.0459 0.0074	0.0069 0.0049	0.0017 0.0018	0.0011 0.0005
	F1-score	0.0006 0.0011	0.0172 0.0068	0.0182 0.0028	0.0040 0.0015
SW-620	AUROC	0.0033 0.0008	0.0020 0.0033	0.0088 0.0266	0.0002 0.0059
	AUPRC	0.0023 0.0023	0.0100 0.0025	0.0010 0.0172	0.0052 0.0036
	F1-score	0.0020 0.0033	0.0075 0.0064	0.0075 0.0033	0.0011 0.0049
NCI-H23	AUROC	0.0081 0.0101	0.0102 0.0040	0.0068 0.0205	0.0015 0.0034
	AUPRC	0.0413 0.0409	0.0221 0.0075	0.0064 0.0223	0.0022 0.0018
	F1-score	0.0260 0.0235	0.0089 0.0023	0.0038 0.0010	0.0001 0.0028
OVCAR-8	AUROC	0.0117 0.0044	0.0016 0.0051	0.0110 0.0013	0.0017 0.0093
	AUPRC	0.0375 0.0405	0.0028 0.0016	0.0007 0.0042	0.0035 0.0153
	F1-score	0.0155 0.0301	0.0006 0.0070	0.0037 0.0024	0.0012 0.0028

to examine the robustness of our framework under an extremely limited auxiliary data scenario.

As shown in Table 24, pseudo-labeling only the validation set leads to a reduction in performance gains compared to our original setting. This behavior indicates that having access to more data for pseudo-labeling can more fully reveal the strength of our framework. At the same time, the results also demonstrate that even when supplied with very limited auxiliary data, our method can still extract meaningful supervisory signals and deliver performance improvements. This resilience under constrained conditions highlights the superiority of our framework, further confirming that FracAug remains effective even in extreme data augmentation settings.

M Original Loss for Baselines

We conduct an additional set of experiments in which baseline models were evaluated under the original, unweighted loss functions,

without applying any rebalancing techniques or class frequency adjustments. The results are summarized in Table 25. As shown, most baseline models experience a substantial decline in performance when trained with their original loss in the imbalanced setting. This degradation is consistent across metrics and datasets, confirming that the performance of these models is sensitive to skewed class distributions. Importantly, this finding validates our experimental choice in the main paper, i.e., the weighted loss was introduced precisely to prevent the baselines from collapsing under severe imbalance, thereby ensuring a fair and meaningful comparison with our proposed method. Without such weighting, the baselines fail to capture minority class patterns effectively, leading to inflated majority class predictions and weakened anomaly or minority detection capability. These newly added results therefore reinforce the significance of using balanced training strategies and further highlight the robustness advantage of our approach.

Table 22: Standard Deviation of AUROC, AUPRC, and F1-score for Table 19.

Datasets	Metrics	MAAv	NS	SM	+FA	GLAv	GLA	+FA	GMv	GM	+FA	FGWMv	FGWM	+FA
MCF-7	AUROC	0.0069	0.0062	0.0154	0.0003	0.0069	0.0208	0.0091	0.0216	0.0173	0.0015	0.0091	0.0427	0.0014
	AUPRC	0.0070	0.0094	0.0107	0.0016	0.0112	0.0320	0.0126	0.0226	0.0272	0.0018	0.0004	0.0854	0.0152
	F1-score	0.0169	0.0018	0.0168	0.0014	0.0003	0.0070	0.0082	0.0144	0.0016	0.0006	0.0330	0.0096	0.0055
MOLT-4	AUROC	0.0080	0.0225	0.0059	0.0065	0.0135	0.0029	0.0004	0.0038	0.0058	0.0001	0.0057	0.0083	0.0146
	AUPRC	0.0110	0.0284	0.0313	0.0016	0.0206	0.0047	0.0013	0.0010	0.0105	0.0003	0.0096	0.0409	0.0116
	F1-score	0.0076	0.0174	0.0110	0.0046	0.0042	0.0030	0.0017	0.0065	0.0013	0.0001	0.0041	0.0136	0.0148
PC-3	AUROC	0.0062	0.0181	0.0296	0.0104	0.0046	0.0195	0.0053	0.0144	0.0179	0.0009	0.0246	0.0141	0.0011
	AUPRC	0.0078	0.0201	0.0329	0.0120	0.0091	0.0277	0.0099	0.0065	0.0290	0.0174	0.0651	0.0373	0.0017
	F1-score	0.0033	0.0003	0.0430	0.0058	0.0111	0.0072	0.0153	0.0235	0.0053	0.0153	0.0163	0.0095	0.0001
SW-620	AUROC	0.0034	0.0043	0.0172	0.0020	0.0061	0.0135	0.0057	0.0029	0.0038	0.0008	0.0302	0.0241	0.0069
	AUPRC	0.0001	0.0059	0.0093	0.0031	0.0151	0.0141	0.0035	0.0191	0.0136	0.0002	0.0645	0.0704	0.0091
	F1-score	0.0004	0.0027	0.0276	0.0019	0.0091	0.0182	0.0052	0.0142	0.0066	0.0013	0.0335	0.0169	0.0071
NCI-H23	AUROC	0.0139	0.0330	0.0020	0.0013	0.0057	0.0077	0.0061	0.0317	0.0018	0.0002	0.0551	0.0073	0.0045
	AUPRC	0.0125	0.0355	0.0049	0.0008	0.0118	0.0076	0.0085	0.0401	0.0070	0.0069	0.0661	0.0037	0.0071
	F1-score	0.0059	0.0175	0.0029	0.0006	0.0100	0.0032	0.0037	0.0083	0.0036	0.0071	0.0305	0.0074	0.0205
OVCAR-8	AUROC	0.0031	0.0111	0.0193	0.0036	0.0023	0.0230	0.0007	0.0169	0.0099	0.0006	0.0194	0.0144	0.0063
	AUPRC	0.0171	0.0175	0.0139	0.0031	0.0083	0.0335	0.0008	0.0361	0.0016	0.0047	0.0783	0.0284	0.0034
	F1-score	0.0221	0.0028	0.0255	0.0013	0.0072	0.0062	0.0004	0.0098	0.0131	0.0035	0.1100	0.0049	0.0072

Table 23: Ablation study.

Datasets	Metrics	GIN	+FA	w/o largest	w/o smallest	w/o WDML	w/o MVP
MCF-7	AUROC	0.5867	0.5976	0.5848	0.5889	0.5860	0.5835
	AUPRC	0.2830	0.2971	0.2842	0.2882	0.2813	0.2790
	F1-score	0.5366	0.5421	0.5317	0.5351	0.5372	0.5350
MOLT-4	AUROC	0.5733	0.5854	0.5770	0.5760	0.5772	0.5754
	AUPRC	0.2830	0.3001	0.2974	0.2862	0.2981	0.2881
	F1-score	0.5072	0.5103	0.5001	0.5085	0.4998	0.5060
PC-3	AUROC	0.5969	0.6119	0.5963	0.6018	0.6026	0.5975
	AUPRC	0.2797	0.2893	0.2797	0.2815	0.2806	0.2780
	F1-score	0.5063	0.5205	0.5055	0.5124	0.5145	0.5092
SW-620	AUROC	0.5938	0.6004	0.5941	0.5938	0.5949	0.5930
	AUPRC	0.2776	0.2813	0.2737	0.2778	0.2800	0.2697
	F1-score	0.5090	0.5155	0.5132	0.5089	0.5082	0.5155
NCI-H23	AUROC	0.5897	0.5968	0.5893	0.5911	0.5866	0.5925
	AUPRC	0.2566	0.2659	0.2564	0.2633	0.2614	0.2656
	F1-score	0.5059	0.5073	0.5054	0.5013	0.4968	0.5013
OVCAR-8	AUROC	0.5935	0.5963	0.5911	0.5918	0.5911	0.5904
	AUPRC	0.2573	0.2612	0.2579	0.2565	0.2573	0.2605
	F1-score	0.5118	0.5123	0.5074	0.5100	0.5081	0.5038

Table 24: Average AUROC, AUPRC, and F1-score on 3 datasets with multiple runs, where the "+FA" represents our original setting, and onlyval represents pseudo-labeling only the validation set.

Datasets	Metrics	GAT	+FA	onlyval	GIN	+FA	onlyval	iGAD	+FA	onlyval	MAAv	+FA	onlyval
UACC257	AUROC	0.5890	0.6174	0.6021	0.5877	0.6015	0.5946	0.5697	0.5748	0.5721	0.5623	0.5947	0.5731
	AUPRC	0.3493	0.3389	0.3250	0.2480	0.2598	0.2631	0.1906	0.1970	0.1956	0.1805	0.2210	0.1847
	F1-score	0.4031	0.4455	0.4491	0.4906	0.4983	0.4908	0.5201	0.5224	0.5214	0.5541	0.5784	0.5582
PROTEINS_full	AUROC	0.6157	0.6836	0.6633	0.5799	0.6174	0.6015	0.5976	0.6206	0.6125	0.6009	0.6217	0.6147
	AUPRC	0.6350	0.7005	0.6933	0.6259	0.6358	0.6261	0.6200	0.6333	0.6271	0.6183	0.6366	0.6357
	F1-score	0.6158	0.6859	0.6508	0.5679	0.6175	0.6024	0.5960	0.6211	0.6137	0.6015	0.6214	0.614
DBLP_v1	AUROC	0.6119	0.6885	0.6489	0.6231	0.8044	0.7844	0.7755	0.7909	0.7901	0.6446	0.6822	0.6712
	AUPRC	0.7507	0.7796	0.7768	0.7201	0.8626	0.8446	0.8377	0.8473	0.8466	0.7689	0.7816	0.7814
	F1-score	0.5782	0.6868	0.6311	0.5996	0.8028	0.7837	0.7749	0.7910	0.7902	0.6252	0.6778	0.6621

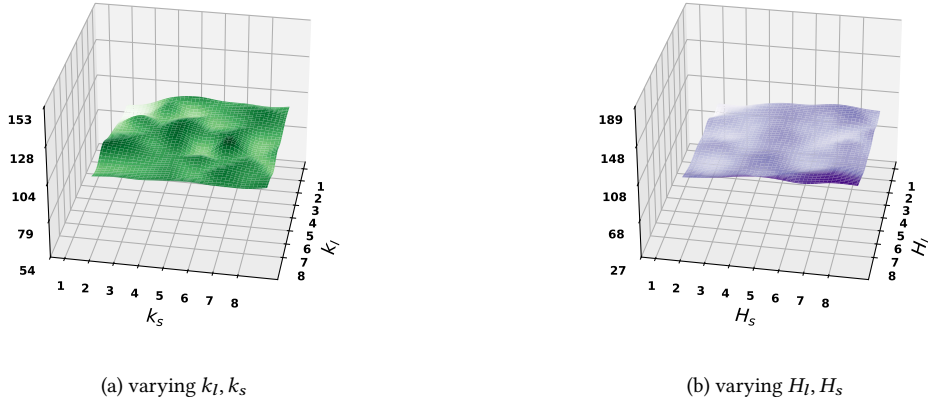
N Running Time when Varying Hyperparameters

Because FGG involves four hyperparameters, k_l , k_s , H_l , H_s , that might influence the running time, we evaluate the computational

overhead under different configurations. We present two 3D plots that report the runtime when varying hyperparameters, as shown in Figures 7. The first plot sweeps k_l and k_s from 1 to 8 and records the training time. The second plot similarly varies H_l and H_s from 1 to 8 and measures the runtime during training.

Table 25: Average AUROC, AUPRC, and F1-score on 3 datasets with multiple runs, where the "+FA" represents our original setting, and oriloss represents the original setting of baselines.

Datasets	Metrics	GAT +FA	oriloss	GIN +FA	oriloss	MAA +FA	oriloss	GM +FA	oriloss
UACC257	AUROC	0.5890	0.6174	0.5583	0.5877	0.6015	0.5749	0.5023	0.5947
	AUPRC	0.3493	0.3389	0.3106	0.2480	0.2598	0.2448	0.0559	0.2210
	F1-score	0.4031	0.4455	0.4039	0.4906	0.4983	0.4772	0.5005	0.5784
PROTEINS_full	AUROC	0.6157	0.6836	0.5739	0.5799	0.6174	0.5517	0.6083	0.6217
	AUPRC	0.6350	0.7005	0.6097	0.6259	0.6358	0.5824	0.6294	0.6366
	F1-score	0.6158	0.6859	0.5695	0.5679	0.6175	0.5503	0.6039	0.6214
DBLP_v1	AUROC	0.6119	0.6885	0.6263	0.6231	0.8044	0.6151	0.6608	0.6822
	AUPRC	0.7507	0.7796	0.7168	0.7201	0.8626	0.7154	0.7868	0.7816
	F1-score	0.5782	0.6868	0.6199	0.5996	0.8028	0.6147	0.6408	0.6778

**Figure 7: Training time when varying k_l, k_s, H_l, H_s for UACC257 based on GIN.**

Across both experiments, we observe that the runtime does increase as the hyperparameters grow, but the increase is small and progresses gradually. Larger k_l and k_s require computing additional eigenvalues, and larger H_l and H_s apply more repeated or deeper fractional operations. Despite this, the overall time remains low and exhibits no sharp growth. These results confirm that FGG’s computational cost scales smoothly with its hyperparameters, and even at higher settings, it maintains an efficient and manageable runtime suitable for practical graph-level anomaly detection.

O Replacement for FGG/MVP

To further evaluate the contribution of FracAug’s main components, we conduct an additional ablation study by replacing FGG. We substitute FGG with two standard perturbation-based augmentations, Node Drop and Edge Drop. As shown in Table 26, these replacements often result in degraded performance compared to our original setting, and sometimes even lower than their baselines. Although our MVP can rule out the noise from non-semantic-preserving generation by mutual verification to some extent, the fail to leverage guaranteed semantic-preserving properties prevents existing data augmentation methods from achieving optimal results, which further demonstrates the effectiveness of our proposed components.

We further replace MVP with a soft ensemble pseudo-labeling strategy that averages the predictions of the original and synthetic samples. This relaxation produces noisier and less reliable pseudo-labels, since it lacks the strict consistency constraint enforced by mutual verification. Consequently, as shown in Table 26, performance becomes unstable or deteriorates across datasets. In contrast, the full FracAug framework, using FGG for high-fidelity synthetic generation and MVP for robust pseudo-label filtering, consistently delivers superior results, underscoring the critical importance of both design components.

Instead of conducting experiments on GIN, we also conduct the same experiments on different backbones to prove that the effectiveness of our framework is a general advantage.

In conclusion, the additional ablation study, where FGG is replaced with either Node Drop or Edge Drop, and MVP is replaced with Soft Ensemble, further reinforces the effectiveness of each individual component. These controlled substitutions consistently lead to inferior performance, thereby validating that all the components contribute uniquely and substantially to the overall design.

P Influence of Middle Eigenvalues

Originally, FGG is intentionally constructed around the largest top- k_l and smallest top- k_s eigenvalues, as these parts of the spectrum encode global structural patterns and localized irregularities, both

Table 26: Ablation study by replacing FGG by Node Drop(ND) or Edge Drop (ED), and replacing MVP by Soft Ensemble (SE).

Datasets	Metrics	GIN	+FA	ND	ED	SE	NS	+FA	ND	ED	SE	GMixup	+FA	ND	ED	SE
MCF-7	AUROC	0.5867	0.5976	0.5878	0.5838	0.5917	0.5496	0.5695	0.5575	0.5442	0.5594	0.5730	0.5935	0.5758	0.5720	0.5738
	AUPRC	0.2830	0.2971	0.2834	0.2832	0.2959	0.2346	0.2455	0.2224	0.2054	0.2248	0.2767	0.3081	0.2941	0.2947	0.2762
	F1-score	0.5366	0.5421	0.5384	0.5309	0.5317	0.5179	0.5721	0.5460	0.5506	0.5589	0.5186	0.5220	0.5070	0.5009	0.5203
MOLT-4	AUROC	0.5733	0.5854	0.5790	0.5774	0.5789	0.5506	0.5663	0.5496	0.5521	0.5557	0.5637	0.5771	0.5737	0.5707	0.5714
	AUPRC	0.2830	0.3001	0.2967	0.2929	0.2949	0.2203	0.2356	0.2271	0.2261	0.2240	0.2411	0.2559	0.2414	0.2545	0.2459
	F1-score	0.5072	0.5103	0.5024	0.5047	0.5101	0.5595	0.5658	0.5606	0.5622	0.5632	0.5306	0.5416	0.5378	0.5300	0.5414
PC-3	AUROC	0.5969	0.6119	0.6036	0.5991	0.6015	0.5688	0.5821	0.5664	0.5639	0.5679	0.5705	0.5782	0.5677	0.5700	0.5731
	AUPRC	0.2797	0.2893	0.2855	0.2837	0.2778	0.2112	0.2385	0.2132	0.2103	0.2149	0.3506	0.3587	0.3473	0.3527	0.3546
	F1-score	0.5063	0.5205	0.5110	0.5058	0.5156	0.5698	0.5848	0.5716	0.5700	0.5725	0.4085	0.4101	0.4083	0.4038	0.4096
SW-620	AUROC	0.5938	0.6004	0.5919	0.5958	0.5932	0.5577	0.5834	0.5634	0.5610	0.5581	0.5839	0.5987	0.5855	0.5834	0.5872
	AUPRC	0.2776	0.2813	0.2721	0.2793	0.2746	0.2026	0.2279	0.2072	0.2041	0.1991	0.2599	0.2728	0.2574	0.2609	0.2658
	F1-score	0.5090	0.5155	0.5112	0.5101	0.5109	0.5641	0.5676	0.5666	0.5651	0.5625	0.5111	0.5220	0.5169	0.5090	0.5110
NCI-H23	AUROC	0.5897	0.5968	0.5847	0.5850	0.5900	0.5792	0.5979	0.5729	0.5780	0.5909	0.5912	0.6014	0.5911	0.5947	0.5984
	AUPRC	0.2566	0.2659	0.2543	0.2522	0.2626	0.2273	0.2407	0.2255	0.2258	0.2295	0.2587	0.2672	0.2522	0.2665	0.2610
	F1-score	0.5059	0.5073	0.5007	0.5031	0.5004	0.5821	0.5835	0.5816	0.5821	0.5776	0.5058	0.5134	0.5127	0.5036	0.5121
OVCAR-8	AUROC	0.5935	0.5963	0.5926	0.5918	0.5933	0.5507	0.5726	0.5603	0.5561	0.5549	0.5786	0.6024	0.5846	0.5840	0.5827
	AUPRC	0.2573	0.2612	0.2527	0.2584	0.2604	0.1775	0.2132	0.1904	0.1758	0.1740	0.2764	0.3072	0.2916	0.2851	0.2883
	F1-score	0.5118	0.5123	0.5068	0.5081	0.5082	0.5461	0.5749	0.5628	0.5518	0.5359	0.4733	0.4766	0.4680	0.4729	0.4740
P388	AUROC	0.5565	0.5913	0.5662	0.5749	0.5645	0.5500	0.5720	0.5507	0.5700	0.5507	0.5469	0.5647	0.5467	0.5486	0.5484
	AUPRC	0.2850	0.3309	0.2933	0.3026	0.2948	0.1958	0.2229	0.1919	0.2077	0.2015	0.1694	0.1957	0.1821	0.1795	0.1789
	F1-score	0.4468	0.4491	0.4445	0.4439	0.4480	0.5520	0.5746	0.5599	0.5678	0.5625	0.5315	0.5373	0.5313	0.5222	0.5224
SF-295	AUROC	0.5844	0.6076	0.5948	0.5972	0.5998	0.5649	0.5753	0.5636	0.5721	0.5624	0.5665	0.6040	0.5862	0.5841	0.5694
	AUPRC	0.2766	0.2832	0.2673	0.2796	0.2780	0.2114	0.2252	0.2154	0.2072	0.2149	0.2004	0.2509	0.2374	0.2278	0.2090
	F1-score	0.4803	0.5047	0.5016	0.4932	0.4984	0.5736	0.5820	0.5752	0.5717	0.5745	0.5245	0.5371	0.5186	0.5260	0.5198
SN12C	AUROC	0.5995	0.6079	0.5972	0.5959	0.6017	0.5509	0.5795	0.5561	0.5538	0.5690	0.5713	0.5984	0.5882	0.5740	0.5817
	AUPRC	0.2696	0.2746	0.2677	0.2682	0.2695	0.1726	0.2164	0.1710	0.1704	0.1855	0.2163	0.2524	0.2386	0.2150	0.2328
	F1-score	0.5030	0.5110	0.5017	0.4991	0.5065	0.5538	0.5770	0.5490	0.5523	0.5650	0.5136	0.5195	0.5175	0.5185	0.5127
UACC257	AUROC	0.5877	0.6015	0.5905	0.5859	0.5952	0.5623	0.5947	0.5768	0.5679	0.5689	0.5853	0.6209	0.5989	0.5945	0.5951
	AUPRC	0.2480	0.2598	0.2442	0.2553	0.2573	0.1805	0.2210	0.1932	0.1813	0.2112	0.2365	0.2963	0.2405	0.2548	0.2727
	F1-score	0.4906	0.4983	0.4978	0.4818	0.4919	0.5541	0.5784	0.5657	0.5615	0.5794	0.4993	0.4898	0.4883	0.4933	0.4780
PROTEINS_full	AUROC	0.5799	0.6174	0.5808	0.5815	0.5979	0.6009	0.6217	0.6054	0.6039	0.6157	0.5411	0.6097	0.5759	0.5523	0.5554
	AUPRC	0.6259	0.6358	0.6139	0.6075	0.6297	0.6183	0.6366	0.6271	0.6353	0.6304	0.5810	0.6244	0.5997	0.5770	0.5992
	F1-score	0.5679	0.6175	0.5770	0.5801	0.5933	0.6015	0.6214	0.6003	0.5991	0.6137	0.5348	0.6102	0.5753	0.5520	0.5494
DBLP_v1	AUROC	0.6231	0.8044	0.7034	0.6791	0.7849	0.6446	0.6822	0.6601	0.6414	0.6712	0.7939	0.7994	0.7891	0.7803	0.7812
	AUPRC	0.7201	0.8626	0.8003	0.7738	0.8411	0.7689	0.7816	0.7794	0.7329	0.7614	0.8503	0.8563	0.8479	0.8394	0.8403
	F1-score	0.5996	0.8028	0.7009	0.6775	0.7847	0.6252	0.6778	0.6467	0.6399	0.6712	0.7937	0.7985	0.7885	0.7805	0.7812

of which are crucial for effectively distinguishing anomalous graphs. To further examine this design choice, we conduct an analysis to determine whether incorporating the middle region of the eigenvalue spectrum offers any meaningful contribution to our fractional augmentation process.

As shown in Table 27, our results indicate that including all eigenvalues does not lead to any noticeable performance improvement across the evaluated datasets. Moreover, extracting these intermediate spectral components requires performing a full eigendecomposition, which incurs a computational complexity of $O(n^3)$, where n denotes the number of nodes in the graph. This stands in sharp contrast to the top- k approximate eigensolvers used in our original design, whose complexity is only $O(k \cdot nnz \cdot t)$, with nnz representing the number of non-zero entries in the adjacency matrix and t denoting the iteration count of the eigendecomposition. Given the minimal performance gains and the substantially higher computational burden, these findings confirm that focusing solely on the largest top- k_l and smallest top- k_s eigenvalues is both an effective and efficient choice for FGG's augmentation strategy.

Q Comparison with warmup models

We conduct an additional experiment to further demonstrate the effectiveness of our proposed model. As described in Section 4, before integrating FracAug into the baseline models, we ensure that the baselines first acquire a foundational understanding of the graph-level anomaly detection dataset by warmup. This step is

crucial to stabilize their predictions and enhance the reliability of pseudo-labeling.

As shown in Table 28, the baseline models that undergo only a brief warmup phase, without sufficient training epochs, are outperformed by their fully trained counterparts. However, once FracAug is integrated into the training process of warmup baselines and then trained for an equivalent number of epochs as the fully-trained baselines, it becomes evident that the warmup models with FracAug achieve substantial performance improvements over the baselines. Ultimately, these models reach state-of-the-art performance, providing strong evidence of the effectiveness of our proposed FracAug framework.

R Comparison with unsuervised and finetuned models

We first conduct comparisons between unsupervised graph-level anomaly detection models from a novel benchmark paper [15] and the FracAug-enhanced GIN. The results in Table 29 show that unsupervised training falls significantly behind our method, even though we only utilize 1% examples as the training set, indicating that label information plays a crucial role in achieving strong anomaly detection performance. Consistent with our earlier experiments in Appendix G, increasing the amount of available labeled data leads to further improvements, reinforcing the importance of label supervision. These findings highlight the importance and effectiveness of our proposed plug-in data augmentation framework, since

Table 27: Ablation study for middle eigenvalues.

Datasets	Metrics	GIN	+FA	w/ middle
MCF-7	AUROC	0.5867	0.5976	0.5958
	AUPRC	0.2830	0.2971	0.2913
	F1-score	0.5366	0.5421	0.5452
MOLT-4	AUROC	0.5733	0.5854	0.5851
	AUPRC	0.2830	0.3001	0.3034
	F1-score	0.5072	0.5103	0.5066
PC-3	AUROC	0.5969	0.6119	0.6182
	AUPRC	0.2797	0.2893	0.2898
	F1-score	0.5063	0.5205	0.5283
SW-620	AUROC	0.5938	0.6004	0.5980
	AUPRC	0.2776	0.2813	0.2820
	F1-score	0.5090	0.5155	0.5110
NCI-H23	AUROC	0.5897	0.5968	0.5923
	AUPRC	0.2566	0.2659	0.2614
	F1-score	0.5059	0.5073	0.5063
OVCAR-8	AUROC	0.5935	0.5963	0.6026
	AUPRC	0.2573	0.2612	0.2617
	F1-score	0.5118	0.5123	0.5119
P388	AUROC	0.5565	0.5913	0.6031
	AUPRC	0.2850	0.3309	0.3376
	F1-score	0.4468	0.4491	0.4483
SF-295	AUROC	0.5844	0.6076	0.6089
	AUPRC	0.2766	0.2832	0.2873
	F1-score	0.4803	0.5047	0.4983
SN12C	AUROC	0.5995	0.6079	0.6082
	AUPRC	0.2696	0.2746	0.2756
	F1-score	0.5030	0.5110	0.5103
UACC257	AUROC	0.5877	0.6015	0.5980
	AUPRC	0.2480	0.2598	0.2525
	F1-score	0.4906	0.4983	0.5006
PROTEINS_full	AUROC	0.5799	0.6174	0.6105
	AUPRC	0.6259	0.6358	0.6321
	F1-score	0.5679	0.6175	0.6096
DBLP_v1	AUROC	0.6231	0.8044	0.8028
	AUPRC	0.7201	0.8626	0.8607
	F1-score	0.5996	0.8028	0.8015

Table 28: Average AUROC, AUPRC, and F1-score on 3 datasets with multiple runs, where the "+FA" represents our original setting, and warmup represents the baseline mode after the warmup phase.

Datasets	Metrics	GAT	+FA	warmup	GIN	+FA	warmup	iGAD	+FA	warmup	MAAv	+FA	warmup
UACC257	AUROC	0.5890	0.6174	0.5568	0.5877	0.6015	0.5606	0.5697	0.5748	0.5590	0.5623	0.5947	0.5499
	AUPRC	0.3493	0.3389	0.3139	0.2480	0.2598	0.1994	0.1906	0.1970	0.1780	0.1805	0.2210	0.1515
	F1-score	0.4031	0.4455	0.3993	0.4906	0.4983	0.4972	0.5201	0.5224	0.5145	0.5541	0.5784	0.5280
PROTEINS_full	AUROC	0.6157	0.6836	0.6012	0.5799	0.6174	0.5723	0.5976	0.6206	0.5631	0.6009	0.6217	0.5875
	AUPRC	0.6350	0.7005	0.6112	0.6259	0.6358	0.5987	0.6200	0.6333	0.6234	0.6183	0.6366	0.6287
	F1-score	0.6158	0.6859	0.6014	0.5679	0.6175	0.5713	0.5960	0.6211	0.5669	0.6015	0.6214	0.5619
DBLP_v1	AUROC	0.6119	0.6885	0.5436	0.6231	0.8044	0.6098	0.7755	0.7909	0.7482	0.6446	0.6822	0.6414
	AUPRC	0.7507	0.7796	0.7207	0.7201	0.8626	0.7014	0.8377	0.8473	0.8272	0.7689	0.7816	0.7329
	F1-score	0.5782	0.6868	0.5425	0.5996	0.8028	0.5910	0.7749	0.7910	0.7447	0.6252	0.6778	0.6239

in real deployment, anomaly detection works usually suffer from the limited label supervision issue.

Furthermore, we compare our performance with Mole-BERT [18], which is a model pretrained in an unsupervised manner and subsequently finetuned with labeled data for a specific domain. As shown in Table 29, the unsupervised pretraining alone produces

suboptimal results, consistent with the behavior of the unsupervised baselines. Even after finetuning with the same labeled data budget as our framework, the performance remains inferior to ours to a large extent. This observation suggests two key points: (1) finetuning may inherit a suboptimal parameter space induced by unsupervised pretraining, limiting its adaptability, and (2) sufficient

Table 29: Comparison with unsupervised and finetuned models. "+FA" represents FracAug-enhanced GIN. "OOM" represents the out-of-memory issue.

Datasets	Metrics	Unsupervised						Mole-BERT		ours	
		OCGIN	GLADC	GLocaKD	OCGTL	SIGNET	CVTGAD	pretrain	finetune	GIN	+FA
PROTEINS_full	AUROC	0.3435	0.3206	0.5596	0.5971	0.5841	0.2485	0.4414	0.5124	0.5799	0.6174
	AUPRC	0.3132	0.3032	0.4941	0.4302	0.4962	0.2815	0.5968	0.5988	0.6259	0.6358
	F1-score	0.2880	0.2879	0.3733	0.2879	0.2879	0.2879	0.3697	0.4882	0.5679	0.6175
DBLP_v1	AUROC	0.4536	0.5340	0.4370	0.5425	0.5131	OOM	0.5077	0.6677	0.6231	0.8044
	AUPRC	0.4651	0.5470	0.4522	0.5552	0.5332	OOM	0.6506	0.7305	0.7201	0.8626
	F1-score	0.3290	0.3288	0.3288	0.3378	0.3378	OOM	0.5069	0.6678	0.5996	0.8028

Table 30: Embedding-distribution comparison results.

Datasets	GCN_n	GCN_a	$SAGE_n$	$SAGE_a$	GAT_n	GAT_a
PROTEINS_full	0.0640	0.0875	0.0334	0.1042	0.0286	0.0858
DBLP_v1	0.0301	0.1034	0.0786	0.0509	0.0521	0.0504

and properly utilized label information is essential for achieving robust performance. Together, these findings further validate the novelty and effectiveness of our plug-in data augmentation framework, which can effectively generate more fidelity label information, even under extremely limited supervision.

S Additional Hyperparameter Analysis

Beyond the results in Section 5.4, we also conduct hyperparameter analysis on DBLP_v1, as shown in Figure 8. The experiment is conducted under the same setting as the one in Section 5.4. We can draw a similar conclusion that all metrics remain stable across this range, demonstrating that FracAug’s pseudo-labeling is robust to threshold selection.

T Embedding-distribution Comparison

We report quantitative results that reflect the embedding-distribution comparison, as shown in 30. Specifically, let d_n (d_a) denote the distance between the original and synthetic graphs in the normal (abnormal) class, and let d denote the distance between the synthetic normal and synthetic abnormal graphs. For a baseline model M , we define

$$M_n = \frac{d_n}{d}, \quad M_a = \frac{d_a}{d}.$$

A smaller value of M_n or M_a is preferred, because it means that the synthetic graphs stay close to the original graphs of the same class, while the two synthetic classes remain well separated. In other words, the within-class discrepancy is small relative to the between-class separation. Therefore, the small ratios reported above indicate that the generated graphs preserve the class-wise embedding distribution well and are consistent with the original graphs.

U Scalability

To address the concern of scalability, we conducted new experiments on the public graph-level anomaly datasets with the largest per-graph sizes that we could identify. These new datasets therefore stress per-graph computational cost more directly than those in the main evaluation, as shown in Table 31. We compare FracAug with

representative graph augmentation baselines, including NodeSam, SubMix, and FGWM. The time and memory are reported in the same way as in Section 5, where we provide the memory and time analysis: for FracAug-enhanced methods, the two terms denote the preprocessing and the FracAug cost, while single-term baselines report their total cost. The results show that FracAug remains effective while keeping competitive runtime and memory.

The results are presented in Table 32. On REDDIT-BINARY, each graph has on average 429.63 nodes and 497.75 edges, which is much larger per graph than the datasets in the main evaluation. FracAug substantially outperforms NodeSam, SubMix, and FGWM in all metrics while maintaining practical runtime and memory. On github_stargazers, whose graphs also have larger average node/edge counts than the main molecular benchmarks, FracAug remains competitive and improves over the corresponding vanilla augmentation baselines, while requiring less runtime than NodeSam/SubMix and much less memory than several baselines.

Algorithmically, FracAug avoids full eigendecomposition: it uses Arnoldi-based top-k eigensolvers and retains only the top k_l largest and top k_s smallest eigenpairs, typically 3 or 4. The preprocessing is performed once and cached, so the heavier spectral step is amortized across training. These new experiments confirm that this design remains practical on the larger datasets, demonstrating the scalability of our method in real-world deployment.

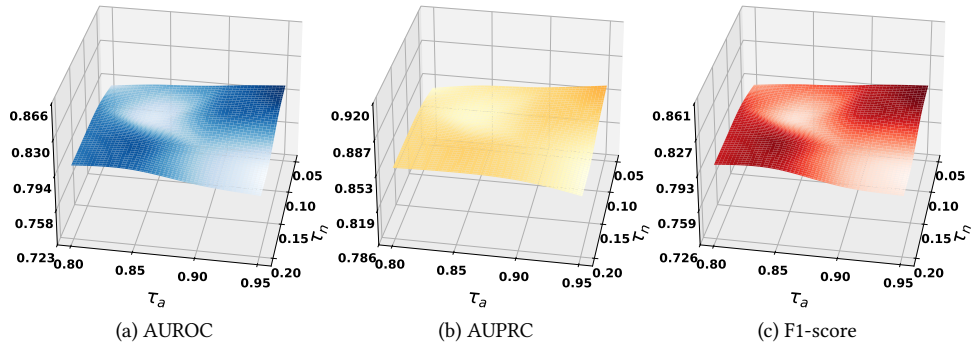
Figure 8: Varying τ_a and τ_n for DBLP_v1 based on GIN.

Table 31: Statistics of scalability datasets.

Dataset	Normal	Abnormal	Abnormal Rate	Avg. Nodes	Avg. Edges	Feature Dim
REDDIT-BINARY	1000	1000	0.5000	429.63	497.75	32
github_stargazers	6808	5917	0.4650	113.79	234.64	30

Table 32: Comparison of scalability datasets.

Dataset	Metric	NSv+FA	NodeSam	SubMix	FGWMv+FA	FGWM
REDDIT-BINARY	AUROC	0.7423	0.5566	0.6148	0.6648	0.5750
	AUPRC	0.8087	0.7416	0.7599	0.7779	0.6863
	F1-score	0.7412	0.4978	0.5828	0.6483	0.5747
	Time	224.21+573.79	433.29	321.68	224.21+604.27	3876.28
	Memory	118.73+360.66	678.58	917.90	118.73+350.59	644.71
github_stargazers	AUROC	0.5230	0.5073	0.5074	0.5776	0.5381
	AUPRC	0.6646	0.6433	0.6405	0.7570	0.6869
	F1-score	0.5216	0.5029	0.5070	0.5535	0.5381
	Time	465.07+315.43	1606.48	1028.63	465.07+321.45	1171.99
	Memory	161.67+542.59	700.80	787.42	161.67+303.37	720.18